

Eigenvalues, eigenvectors

Standard method:

- eigenvalues: roots of the characteristic polynomials.

Difficult problem!

- eigenvectors: solution of a singular set of linear equations.

Easy.

Power method: find the largest eigen value

Let $\underline{A}$ be an n diagonalizable matrix with

eigenvalues such that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$

and with the corresponding eigenvectors

$\underline{v}_i$. We know, that $\underline{v}_i$ span the whole space

- if $\underline{A}$ is diagonalizable

Let us start from an arbitrary vector

$$\underline{b}_0 = c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_n \underline{v}_n, \text{ with } c_1 \neq 0$$

Apply the operator $\underline{A}$ to $\underline{b}_0$

$$\underline{A} \underline{b}_0 = \underline{A} \sum_i c_i \underline{v}_i = \sum_i c_i \underline{A} \underline{v}_i = \sum_i c_i \lambda_i \underline{v}_i$$

$$\underline{A}^k \underline{b}_0 = \sum_i c_i \lambda_i^k \underline{v}_i$$

12

if k is sufficiently large then

$$|\lambda_1|^n \gg |\lambda_i|^n \quad i > 1$$

So $\lim_{n \rightarrow \infty} \frac{A^n}{n} \underline{b}_0 = \lambda_1^n C_1 \underline{v}_1 = \lambda_1 \underline{v}_1$ converges to the

eigenvector of the largest eigenvalue,

obviously we can get λ_1 by $\underline{A} \underline{x} \cdot \underline{v}_1 = \lambda_1 \underline{x} \cdot \underline{v}_1$

The practical method includes a norma-

lization:

$$\underline{b}^{(k+1)} = \frac{\underline{A} \underline{b}^{(k)}}{\|\underline{A} \underline{b}^{(k)}\|}$$

$\underline{b}_0$ has to be chosen carefully, or
two perpendicular starting points.

Eigenvalue shift:

We can manipulate the eigenvalues of
the original matrix:

$$- \underline{A} \rightarrow \underline{A}^{-1} : \lambda_1, \dots, \lambda_n \rightarrow \frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

$$- \underline{A} \rightarrow \underline{A} + \underline{I} \cdot \alpha : \lambda_1, \dots, \lambda_n \rightarrow \lambda_1 + \alpha, \dots, \lambda_n + \alpha$$

We can manipulate the largest eigenvalue.

- General procedure: Shift close to the eigenvalue, then use the power method on the inverse. Useful if the approximate position is known

- Second largest eigenvalue with $\|v_1\| = 1$

$$A \rightarrow A - \lambda_1 v_1 v_1^T$$

$v_1 v_1^T = v_1 \cdot v_1$ outer product has two

different eigenvalues $\lambda_1 = \|v_1\|^2 = 1$ and

$$\lambda_2 = \dots = \lambda_n = 0$$

Numerical derivation

The definition of the derivative of the function $f(x)$ in the point a is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

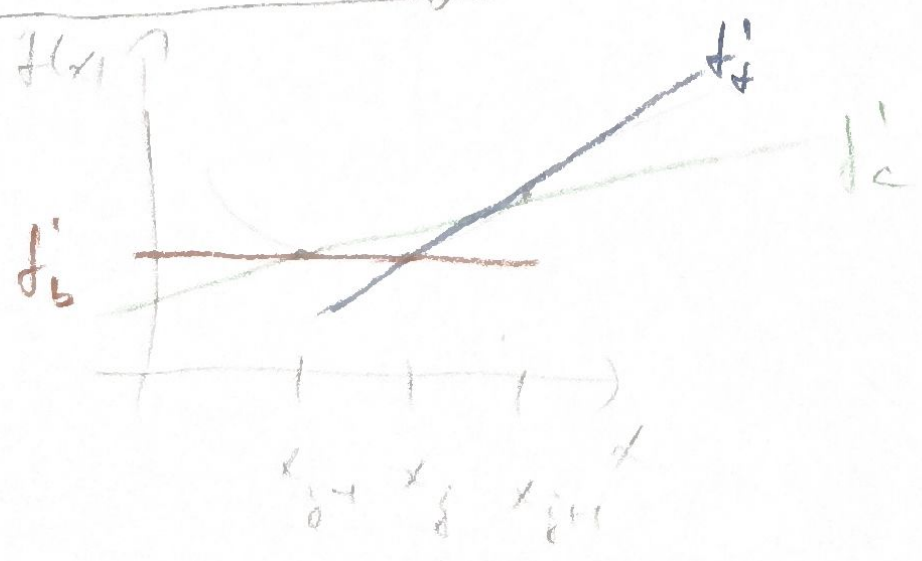
Generally $f(x)$ is known at discrete points

$$x_j \quad j = 1, 2, \dots, n \quad x_{j+1} - x_j = h$$

Forward difference: $f'_f(x_j) = \frac{f(x_{j+1}) - f(x_j)}{x_{j+1} - x_j} = \frac{f(x_{j+1}) - f(x_j)}{h}$

Backward difference: $f'_b(x_j) = \frac{f(x_j) - f(x_{j-1})}{h}$

Central difference: $f'_c(x_j) = \frac{f(x_{j+1}) - f(x_{j-1}))}{2h}$



Error:

Start from the Taylor series

$$f(x) = \frac{f(x_j)(x-x_j)^0}{0!} + \frac{f'(x_j)(x-x_j)^1}{1!} + \frac{f''(x_j)(x-x_j)^2}{2!} + \dots$$

- Forward difference

$$x = x_{j+1}, \text{ again } x_{j+1} - x_j = h$$

$$f(x_{j+1}) = f(x_j) + \underbrace{f'(x_j)(x_{j+1} - x_j)}_h + \frac{1}{2}f''(x_j)(x_{j+1} - x_j)^2 + \dots$$

$$f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + \left(-\frac{1}{2}f''(x_j)h - O(h^2) \right)$$

$$f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + O(h)$$

The error of the forward difference is h .

- Backward difference

Similarly the error is proportional to h .

Central difference: Use the Taylor series at two points

$$f(x_{j+1}) = f(x_j) + f'(x_j)h + \frac{1}{2}f''(x_j)h^2 + O(h^3)$$

$$f(x_{j-1}) = f(x_j) - f'(x_j)h + \frac{1}{2}f''(x_j)h^2 + O(h^3)$$

$$f(x_{j+1}) - f(x_{j-1}) = 2f'(x_j)h + 0 + O(h^3)$$

$$f'(x_j) = \frac{f(x_{j+1}) - f(x_{j-1}))}{2h} + O(h^2)$$

One can go further and using more distant points cancel higher terms in the series. Eg:

$$f'(x_j) = \frac{f(x_{j-2}) - 8f(x_{j-1}) + 8f(x_{j+1}) - f(x_{j+2}))}{12h} + O(h^4)$$

Important: numerical precision may limit the gain

